KPPSC PMS • GEOGRAPHY (PAPER-II)

EXECUTIVE MASTER EXAMINATION NOTES • Q.5: Overpopulation Threats & Demographic Pressures in Pakistan: Resource Strain & Youth Bulge

Target Examination	KPPSC PMS (Provincial Management Service) — Khyber Pakhtunkhwa
Paper & Subject	Geography Paper-II (Human, Economic & Regional Geography — 100 Marks)
EXAM QUESTIONS LINKED:	• 2025 Q.3: "Analyze the major threats posed by overpopulation at global and national scales. Critically examine the demographic pressures facing Pakistan and their implications for economic development, resource management, and social stability. (20 Marks)" • 2018 Q.4: "What are the factors responsible for the uneven distribution of population with special reference to Pakistan? (20 Marks)"
Syllabus Section	Section A & C: Human Geography & Regional Geography of Pakistan — Population growth, resource pressure, urbanization
Allocated Marks & Time	20 Marks | 35 Minutes Exam Writing Allocation
Core Analytical Thesis	Pakistan confronts an existential demographic crisis characterized by a rapid 2.55% intercensal growth rate, culminating in an official population of 241.5 million (7th Population and Housing Census 2023). This demographic momentum exerts catastrophic structural pressure across four critical geographical thresholds: acute per capita water depletion (<900 m^3), severe arable land fragmentation, uncontrolled peri-urban sprawl, and a precarious youth bulge (64% under age 30) that risks metastasizing into severe socio-political volatility unless swiftly capitalized via human capital formation and radical fertility decline.

🎯 EXAM QUESTIONS LINKED

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📌 EXAM ESSENTIAL TAKEAWAY

CENSUS 2023 BASELINE: Pakistan's population reached 241.49 million in the 2023 Digital Census (an increase of ~34 million in merely six years since 2017), growing at a blistering 2.55% annually—the highest growth rate in South Asia.

THE WATER SCARCITY THRESHOLD: Per capita water availability has collapsed from 5,260 m^3 in 1951 to below 900 m^3 in 2023 (well past the internationally recognized water-scarcity threshold of 1,000 m^3), driven primarily by population multiplication over a fixed Indus Basin hydrologic discharge.

THE YOUTH BULGE DOUBLE-EDGED SWORD: 64% of Pakistan's population is under 30 years of age, and 29% is between 15 and 29. With 26.2 million out-of-school children (UNICEF 2023) and high youth unemployment, this demographic dividend threatens to become a demographic disaster.

SPATIAL REORGANIZATION: Massive rural-to-urban migration is transforming cities like Karachi, Lahore, and Peshawar into unmanageable mega-agglomerations characterized by proliferating katchi abadis (informal settlements), groundwater depletion, and acute air pollution (smog).

1. CONCEPTUAL MIND MAP & THEMATIC ROADMAP

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PAKISTAN DEMOGRAPHIC STRESS ARCHITECTURE (CENSUS 2023 ANALYSIS)
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[1. EMPIRICAL TRAJECTORY (1951 - 2023 CENSUSES)]
  * 1951: 33.7 Million  ──►  1972: 65.3 Million  ──►  1998: 132.3 Million  ──►  2017: 207.7 Million  ──►  2023: 241.49
Million.
  * Intercensal Growth Rate (2017-2023): 2.55% per annum (Doubling time: ~27 years).
  * Total Fertility Rate (TFR): ~3.6 births per woman (Contraceptive Prevalence Rate: ~34%).
        │
        ▼ (The Triple Threat: Resource Strain & Planetary Limits)
[2. CORE RESOURCE PRESSURES]
  ├── (A) HYDRIC COLLAPSE (Water Stress)
  │     ├── 1951: 5,260 m³/capita ──► 2023: < 880 m³/capita (Extreme Water Scarcity).
  │     └── Drivers: Fixed Indus Basin annual inflows (~140 MAF) divided by surging denominator.
  ├── (B) EDAPHIC FRAGMENTATION (Food Security)
  │     ├── Arable Land per Capita: Collapsed from 0.45 hectares (1960) to < 0.15 hectares (2023).
  │     └── Phenomenon: Severe land subdivision under inheritance laws; loss of prime agricultural land to housing
schemes.
  └── (C) SPATIAL INFRASTRUCTURAL GRIDLOCK (Hyper-Urbanization)
        ├── Urban Population: Officially ~38.8% (Functionally >50% including peri-urban census tracts).
        └── Crisis: Influx into Karachi, Lahore, Rawalpindi, Peshawar; 45%+ urbanites live in informal slums (katchi
abadis).
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        ▼ (The Socio-Demographic Crucible)
[3. THE YOUTH BULGE PARADOX (Ages 15-29)]
  * Demographic Dividend Opportunity: Massive potential productive workforce, low dependency ratio window.
  * Structural Bottlenecks (The Dividend Deficit):
      ├── Education Crisis: 26.2 million out-of-school children; low technical vocational penetration.
      ├── Labor Market Failure: Informal economy absorbs 72% of workforce; high youth unemployment & underemployment.
      └── Brain Drain: Over 800,000 skilled professionals emigrating annually (Bureau of Emigration 2023).
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        ▼
[4. STRATEGIC POLICY FRAMEWORK FOR STABILIZATION]
  ├── Aggressive Family Planning: Raise CPR from 34% to 60%; integrate Lady Health Workers (LHWs) with digital
outreach.
  ├── Female Educational & Economic Inclusion: Parity in secondary/tertiary education; formal female labor
participation.
  ├── National Water Conservation: Canal lining, drip/sprinkler micro-irrigation, water metering in metropolitan hubs.
  └── Polycentric Urban Planning: Develop intermediate secondary cities to halt primal urban saturation.
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2. ANALYTICAL DIMENSIONS & SOCIO-SPATIAL/THEORETICAL MATRIX

Demographic Dimension	Empirical Baseline (Census 2023 / Official Data)	Critical Geographical & Socio-Economic Impact
National Population Scale	241.49 million (Punjab: 127.7M, Sindh: 55.7M, KP: 40.85M, Balochistan: 14.9M, ICT: 2.36M).	Pakistan is the 5th most populous country globally; demands massive continuous expansion of food, water, and social infrastructure.
Annual Growth & Fertility	Growth rate: 2.55% p.a.; Total Fertility Rate (TFR): 3.6; Contraceptive Prevalence Rate (CPR): 34%.	Adds ~6.1 million people annually (equivalent to adding an entire city of Lahore every two years), cancelling out per capita GDP growth.
Per Capita Water Resource	Collapsed from 5,260 m³ (1951) to <880 m³ (2023); PCRWR projects absolute water exhaustion.	Ranks among the top 10 most water-stressed nations globally; acute inter-provincial disputes over Indus water distribution (IRSA).
Arable Land & Food Security	Arable land per capita <0.15 ha; food grain imports (wheat, pulses, edible oils) exceed $8B annually.	High malnutrition (stunting rate >40% among children under 5); agricultural land lost to speculative housing societies.
Youth Cohort & Education	64% under age 30; 26.2 million out-of-school children; literacy rate stagnant at 58%.	Creates a massive population of uneducated, unemployable youth vulnerable to radicalization, criminality, or forced informal labor.
Urban Housing & Slums	Urbanization at 38.8% (Census); 45%+ urban dwellers reside in informal settlements (katchi abadis).	Acute shortage of municipal sanitation, piped water, public transit; severe seasonal toxic smog crises in Punjab.

3. 20-MARK MASTER MODEL ESSAY (KPPSC PMS BENCHMARK)

Introduction: Pakistan's Demographic Crisis at the Crossroads

Pakistan stands at an existential demographic crossroads. According to the official results of the 7th Population and Housing Census (2023)—the country's first-ever digital census—Pakistan's population has surged to 241.49 million inhabitants, solidifying its position as the fifth most populous country on the planet. With an annual intercensal growth rate of 2.55%, Pakistan is the fastest-growing major nation in South Asia, vastly outstripping India (0.9%), Bangladesh (1.1%), and Sri Lanka (0.5%). This relentless demographic expansion is not merely an abstract statistical dilemma; it represents a profound geographic mismatch between an exponentially multiplying human denominator and a strictly finite, environmentally degraded biophysical resource base. Unless urgent, state-level interventions decouple demographic momentum from resource consumption, Pakistan faces catastrophic socio-ecological destabilization.

Empirical Realities of the 2023 Digital Census: Spatial Distributions and Momentum

The 2023 Census reveals profound spatial variations in demographic momentum across Pakistan's provinces: - Punjab remains the demographic core with 127.68 million inhabitants (52.9% of the national total), growing at 2.53% annually. - Sindh houses 55.69 million people (23.1%), with a growth rate of 2.57%, characterized by intense demographic concentration in the megacity of Karachi (18.8 million). - Khyber Pakhtunkhwa (including the merged tribal districts) has reached 40.85 million people (16.9%), maintaining an annual growth rate of 2.38%. - Balochistan experienced the highest intercensal growth rate at 3.20%, bringing its population to 14.89 million, driven by high total fertility rates, improved digital coverage, and refugee persistence. - The federal capital, Islamabad, surged to 2.36 million, reflecting its status as an administrative and real estate growth magnet. Underpinning this rapid growth is an alarmingly high Total Fertility Rate (TFR) of approximately 3.6 children per woman, far above the replacement fertility level of 2.1. The Contraceptive Prevalence Rate (CPR) remains culturally stagnated at roughly 34%, burdened by low female literacy, patriarchal cultural norms, inadequate public health access in rural peripheries, and the historical failure of state-sponsored family planning programs.

Hydric and Edaphic Collapse: The Threat to Foundational Ecological Capital

The most direct geographic threat posed by overpopulation in Pakistan is the exhaustion of its foundational ecological resources: water and arable soil. - Water Scarcity and the Indus Basin: Pakistan's civilization is entirely dependent on the Indus River Basin, which provides over 90% of the country's agricultural water. While the Indus Basin's surface water inflows are hydrologically fixed (averaging roughly 140 million acre-feet annually, subject to accelerating glacial melt volatility), dividing this fixed volume by a skyrocketing population has caused per capita water availability to collapse. In 1951, Pakistan enjoyed an abundant 5,260 cubic meters per capita per annum. By 2023, this figure dropped below 880 cubic meters, placing Pakistan firmly in the category of 'water-scarce' nations under United Nations standards. The Pakistan Council of Research in Water Resources (PCRWR) warns that Pakistan could reach absolute water exhaustion by 2025–2030. Unregulated tubewell extraction in Punjab and Balochistan has caused water tables to drop by more than one meter per year, depleting non-renewable deep fossil aquifers. - Edaphic Fragmentation and Food Insecurity: Overpopulation has led to the hyper-fragmentation of agricultural landholdings under Islamic inheritance laws, reducing the average farm size to economically unviable subsistence plots (<5 acres). Simultaneously, prime agricultural soils in peri-urban fringes (e.g., around Lahore, Faisalabad, and Gujranwala) are being aggressively paved over by speculative private housing societies, undermining domestic crop yields. Consequently, a nation with one of the world's most extensive contiguous irrigation networks now spends billions of scarce foreign exchange dollars importing staple commodities, including wheat, pulses, and edible oil.

Urban Congestion and Infrastructure Gridlock: The Proliferation of Katchi Abadis

Rapid population growth combined with agrarian distress has triggered an unprecedented rural-to-urban migration exodus. While the 2023 Census officially records the national urban proportion at 38.8%, spatial and satellite analyses demonstrate that more than 50% of the population functionally resides in high-density urban and peri-urban environments. This uncontrolled urban surge has overwhelmed metropolitan infrastructure. In Karachi, Pakistan's commercial gateway, municipal institutions fail to deliver basic civic amenities: over 50% of the city lacks formal piped water access, compelling millions to rely on exploitative private tanker mafias. More than 45% of the national urban population resides in informal settlements ('katchi abadis') characterized by extreme overcrowding, unpaved open

sewers, zero solid waste management, and absence of formal tenure rights. Furthermore, vehicular emissions, brick kilns, and industrial effluent trap cities across the Indus Basin (most notably Lahore and Multan) under toxic blankets of winter smog, causing severe public health emergencies and substantial economic productivity losses.

The Youth Bulge: Demographic Dividend or Socio-Political Powder Keg?

Approximately 64% of Pakistan's population is under 30 years of age, and roughly 29% falls into the pivotal 15–29 age bracket. In theoretical demography, this youthful demographic profile represents a historic window of opportunity—a 'Demographic Dividend'—where a low dependency ratio can catalyze unprecedented economic accumulation if properly harnessed. However, in Pakistan, this dividend is turning into a grave structural liability. According to UNICEF and the Ministry of Federal Education (2023), over 26.2 million children are currently out of school—the second-highest absolute number in the world. The national literacy rate has remained virtually stagnant at 58% (with female literacy languishing below 48%). The formal economy, constrained by chronic macroeconomic instability, fiscal deficits, and low industrialization, cannot generate the two million new formal jobs required annually just to absorb incoming labor entrants. Consequently, millions of undereducated youth drift into low-productivity informal jobs, underemployment, or criminality. Moreover, an unprecedented 'brain drain' has emerged: in 2023 alone, over 800,000 skilled and educated Pakistani professionals emigrated abroad, stripping the country of the vital human capital needed for technological modernization.

Conclusion: Strategic Imperatives for National Demographic Stabilization

In conclusion, overpopulation is not merely one among many developmental challenges facing Pakistan; it is the master crisis that exacerbates every economic, environmental, and geopolitical vulnerability. The country cannot build schools, hospitals, water reservoirs, or power plants fast enough to keep pace with a growth rate of 2.55%. To avert socio-ecological catastrophe, Pakistan must enact an aggressive, multi-pronged national survival strategy: (1) elevate family planning to an emergency national security priority, expanding the Contraceptive Prevalence Rate to at least 60% through community health mobilization; (2) achieve universal female secondary education and provide vocational training to unlock female workforce participation; (3) overhaul water governance by replacing flood irrigation with precision drip systems, enforcing mandatory water metering, and protecting recharge aquifers; and (4) implement strict land-use zoning laws to protect prime agricultural land from residential encroachment. Without decisive demographic stabilization, Pakistan's geographic carrying capacity will be catastrophically breached.

4. SCHOLARLY PERSPECTIVES & GEOGRAPHICAL AXIOMS TABLE

Scholar / Geographer	School of Thought / Paradigm	Quotable Philosophical Thesis / Axiom
Dr. Mahbub ul Haq (1996)	Human Development Paradigm	*"Human development is the end; economic growth a means. When population growth outpaces human capital investment, economic growth becomes an illusion that masks human deprivation."*
Akhtar Hameed Khan (1985)	Rural Development & Social Mobilization	*"The true bottleneck in our development is the rural-urban population tidal wave; without local institutional capacity, urban infrastructure will simply collapse under demographic weight."*
United Nations Population Fund (UNFPA)	Pakistan Demographic Overview (2023)	*"Pakistan's demographic trajectory is unsustainable; adding over 6 million people annually strains the limits of water, food, and social cohesion."*
State Bank of Pakistan (SBP Report)	Economic Advisory Analysis	*"Pakistan's low national savings rate and persistent fiscal deficit are structurally anchored to its high demographic dependency ratio, starving the economy of developmental capital."*
Ishrat Husain (2018)	Institutional Economics & Governance	*"Unless Pakistan captures its youth dividend through rapid skill acquisition and female labor empowerment, the youth bulge will inevitably turn into an explosive demographic liability."*